

# Muhammad Abdullah Butt

## Full Stack Engineer & AI Researcher

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Gujranwala, Pakistan

### PROFESSIONAL SUMMARY

Full Stack Engineer and AI Researcher with 3+ years of production experience building scalable web and mobile platforms and LLM-powered systems. Published author of MAPF-Lite, a peer-reviewed deepfake detection framework achieving 99.34% accuracy at 0.566M parameters, accepted at FLINS-ISKE 2026 and included in a Springer Nature proceedings volume. Shipped 5+ live production products across healthcare, fintech, livestock management, fashion, and SaaS. Deep expertise in RAG pipelines, agentic AI, React Native, and full-stack web development. Certified Oracle Generative AI Professional and IBM Agentic AI Expert.

### RESEARCH PUBLICATIONS

#### MAPF-Lite: A Parameter-Efficient Multimodal Framework for High-Fidelity Deepfake Detection

FLINS-ISKE Journal, 2026 (Published) — Springer Nature Proceedings Volume — Sole architect and lead researcher

- 99.34% accuracy and 99.84% AUC on FakeAVCeleb; 99.40% cross-dataset accuracy on DeepfakeTIMIT (zero-shot)
- Reduced trainable parameters by 99.6% to 0.566M via frozen CLIP/Whisper backbones with SRM and DCT analysis
- 4.7× real-time inference at 1.1 GB peak memory; edge and mobile deployable
- XAI modules with visual heatmaps and frame-level suspicion scores for forensic transparency

#### Deepfake Detection in the Multimodal Era: A Systematic Review

Review Paper (In Preparation) — Lead author

- Surveys multimodal deepfake detection techniques, parameter-efficient architectures, and benchmark datasets, extending MAPF-Lite findings

### WORK EXPERIENCE

#### Senior Full-Stack AI Engineer

Mar 2026 – Present

[Huzzle.com](#) | London, UK (Remote)

- Building AI-driven recruitment infrastructure integrating LLM pipelines with full-stack product features for a live, high-traffic B2C platform

#### Full Stack Engineer

Jan 2023 – Feb 2026

[MAQ Enterprises Ltd.](#) | Rawalpindi, Pakistan

- Engineered production web applications with Next.js and Node.js; reduced page load time by 40% via SSR optimization and code splitting
- Designed RESTful APIs and optimized PostgreSQL and MongoDB schemas, improving data retrieval speed by 35% on critical endpoints
- Implemented Jenkins and Docker CI/CD pipelines, cutting release cycle time by 30%
- Automated end-to-end testing with Cypress and Jest, reducing production bug incidence by 25%
- Mentored 3+ junior developers through code reviews; provisioned infrastructure-as-code using Terraform

#### AI Researcher

2024 – 2026

[GIFT University](#) | Gujranwala, Pakistan

- Designed MAPF-Lite multimodal architecture (CLIP + Whisper + ViT) for deepfake detection, published in peer-reviewed journal and Springer Nature proceedings
- Reduced trainable parameters by 99.6% while maintaining state-of-the-art accuracy through parameter-efficient fine-tuning (PEFT)
- Achieved 30% reduction in detection latency by developing lightweight attention mechanisms for edge deployment

#### Full-Stack Developer

2025 – May 2026

[Estabraq \(Fashion Brand\)](#) | Pakistan

- Architected live fashion e-commerce platform (estabraq.pk) with Next.js App Router, Prisma ORM, and Supabase; achieved sub-2-second load times in production
- Integrated GraphQL APIs, Redis caching, and CDN image optimization, reducing server latency by 30%

- Automated deployment with Docker CI/CD, cutting release cycles by 25%

## KEY PROJECTS

**Pointr — Livestock Management Platform (Web + Mobile)** [livestock.pointrsolutions.co.za](https://livestock.pointrsolutions.co.za) | Expo • React Native • Django • JWT • Android • iOS • EAS

- Built and published a full-stack livestock management ecosystem: web dashboard + Expo / React Native mobile app (Android + iOS) with real-time sync via Django backend and JWT auth
- Engineered offline-first two-phase architecture: data captured in the field without connectivity, uploaded on mobile data — zero data loss across 100% of sync cycles in production
- Integrated XR5000 weigh-scale via Wi-Fi, parsing ADI XML sessions (EID, weight, timestamp, health notes); UUID-keyed uploads eliminate duplicate weigh-ins on retry
- Wrote back clean animal data (EID, breed, weight history) to scale indicator, replacing bare tag numbers with real farm records
- Isolated all device logic behind a single adapter + mock server; ~90% of development done without hardware, reducing debugging cost significantly
- Released via EAS: signed .aab for Google Play with OTA updates for JS-only fixes, bypassing full store review cycles

**Estabraq — Fashion E-Commerce Platform** [estabraq.pk](https://estabraq.pk) | Next.js • TypeScript • Supabase • Prisma ORM • GraphQL • Redis

- Live production platform with sub-2-second load times, real-time inventory, and secure authentication
- GraphQL APIs + Redis caching reduced client-server latency by 30%; CDN image optimization cut bounce rates measurably
- React Native mobile companion app in progress for omnichannel experience

**MyPDFMate — PDF SaaS** [mypdfmate.com](https://mypdfmate.com) | Next.js • FastAPI • AI Processing

- Live production SaaS for PDF processing and AI-powered document workflows

**Khushi Motors — Automotive Platform** [khushimotors.com](https://khushimotors.com) | Next.js • Full Stack

- Live automotive marketplace with full product catalogue, filtering, and inquiry management

**DeployGenius — AI Chatbot SaaS** | FastAPI • LangChain • OpenAI GPT-4 • Next.js 14 • Redis • PostgreSQL • Stripe

- Multi-tenant AI SaaS with RAG grounding, multi-session chat management, and agentic workflows
- Stripe subscription lifecycle integration: tiered pricing, webhooks, secure checkout; optimized API costs by 25%

**LingumedAI — Clinical AI for Nurses** | React • FastAPI • LangChain • RAG • pgvector • OpenAI API

- AI clinical assistant giving nurses real-time access to diagnostic guidance and drug interaction checks, reducing physician dependency in critical moments
- Domain-specific RAG pipeline ingesting medical literature and clinical guidelines into pgvector; hallucination-resistant responses with explicit uncertainty escalation
- Role-based access control and audit logging for healthcare data governance compliance

**ForexAI — Mobile Trading & Prediction App** | React Native • XGBoost • Kronos AI • Python

- Forex education app with AI market prediction; fine-tuned XGBoost + Kronos AI ensemble on historical price data
- In-app learning curriculum paired with live prediction signals for retail traders

## TECHNICAL SKILLS

|                 |                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------|
| AI & LLM        | LangChain · LlamaIndex · OpenAI API · Anthropic Claude · RAG Pipelines · Pinecone · pgvector · Agentic Workflows |
| ML & Fintech AI | XGBoost · Decision Trees · Kronos AI · Model Fine-Tuning · Ensemble Modeling · Time-Series Forecasting           |
| Computer Vision | PyTorch · Vision Transformers · CLIP · Whisper · PEFT · Model Quantization · Edge Deployment                     |
| Frontend        | Next.js · React · React Native · Expo · TypeScript · Tailwind CSS · GraphQL                                      |
| Backend         | FastAPI · Node.js · Django · Prisma ORM · RESTful APIs · Redis                                                   |
| Databases       | PostgreSQL · MongoDB · Supabase                                                                                  |
| DevOps & Cloud  | Docker · Jenkins · Terraform · CI/CD · EAS · Vercel · Railway · Oracle Cloud · AWS                               |

## EDUCATION

**Bachelor of Science in Computer Science**

GIFT University, Gujranwala, Pakistan

Thesis: MAPF-Lite — published at FLINS-ISKE 2026 and included in Springer Nature proceedings volume

Sep 2022 – Mar 2026

CERTIFICATIONS

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|------------------------------------------------------------------|------------------------------------------------------------------|
| <b>OCI Certified Generative AI Professional</b><br>Oracle · 2025 | <b>NLP and Computer Vision Specialist</b><br>IBM                 |
| <b>OCI Certified AI Foundations Associate</b><br>Oracle · 2025   | <b>Generative AI: Language Modeling with Transformers</b><br>IBM |
| <b>Agentic AI Expert</b><br>IBM                                  | <b>Advanced React</b><br>Meta                                    |
| <b>Build RAG Applications</b><br>IBM · 2026                      | <b>Machine Learning Using Python</b><br>IBM                      |